Chun-Hua Tsai, Ph.D.

Personal Data

1110 S. 67th Street, 285A PKI
Omaha, NE, 68182, USA

chunhuatsai@unomaha.edu
http://www.cht77.com

Areas of Specialization

Human-Computer Interaction (HCI); Health Informatics; Recommender Systems;
Social Computing; Intelligent User Interfaces; Explainable Artificial Intelligence (XAI)

Appointments

- **Assistant Professor**, 2021-Present
  Department of Information Systems and Quantitative Analysis (ISQA)
  College of Information Sciences and Technology, University of Nebraska at Omaha
- **Assistant Research Professor**, 2019-2021
  College of Information Sciences and Technology, The Pennsylvania State University
  Mentor: Dr. John M. Carroll

Education

- **Ph.D. in Information Science**, 2013-2019
  School of Computing and Information, University of Pittsburgh
  Advisor: Dr. Peter Brusilovsky
- **Master of Science in Management Information Systems**, 2007-2009
  Department of Management Information Systems, National ChengChi University
  Advisor: Dr. Jiann-Ming Yang
- **Visiting Scholar**, Spring 2009
  Dept. of Business and Management Science, NHH Norwegian School of Economics
- **Bachelor of Business Administration in Information Management**, 2003-2007
  Department of Information Management, Tamkang University

Publications

Journal Papers

[J1]. Gupta, S., Carroll, J. M., & **Tsai, C. H.** (2025). Distance Matters in Citizen-Based Water Quality Monitoring. *ACM on Human-Computer Interaction (CSCW)*. (to appear)

[J2]. **Tsai, C. H.**, Huang, C., Chen, Y. C., Zendejas, E., Krafka, S., & Zendejas, J. (2024). Co-Design Smart Disaster Management Systems with Indigenous Communities. *Digital Government: Research and Practice*. (ACM DGOV). https://doi.org/10.1145/3660643

[J3]. You, Y., **Tsai, C. H.**, Li, Y., Ma, F., Heron, C. & Gui, X. (2023). Beyond Self-diagnosis: How a Chatbot-based Symptom Checker Should Respond. ACM Transactions on Computer-Human Interaction (TOCHI). https://doi.org/10.1145/358995

[J4]. Gui, F., **Tsai, C. H.**, Vajda, A. & Carroll, J. M. (2022) Workout connections: investigating social interactions in online group exercise classes. International Journal of Human-Computer Studies (IJHCS). https://doi.org/10.1016/j.ijhcs.2022.102870

[J5]. Xie, J., Reddie, M., Lee, S., Billah, S., Zhou, Z., **Tsai, C. H.** & Carroll, J. M. (2022). Iterative Design and Prototyping of Computer Vision Mediated Remote Sighted Assistance. ACM Transactions on Computer-Human Interaction (TOCHI). https://doi.org/10.1145/3501298

[J6]. Gupta, S., Jablonski, J., **Tsai, C. H.** & Carroll, J. M. (2022). Instagram of Rivers: Facilitating Distributed Collaboration in Hyperlocal Citizen Science. *ACM on Human-Computer Interaction (CSCW)*. https://doi.org/10.1145/3512944

[J7]. **Tsai, C. H.**, Gui, X., Kou, Y. & Carroll, J. M. (2021). With Help from Afar: Cross-Local Communication in an Online COVID-19 Pandemic Community *ACM on Human-Computer Interaction (CSCW)*. https://doi.org/10.1145/3476062

[J8]. Sandbulte, J., **Tsai, C. H.** & Carroll, J. M. (2021). Working Together in a PhamilySpace: Facilitating Collaboration on Healthy Behaviors Over Distance. *ACM on Human-Computer Interaction (CSCW)*. https://doi.org/10.1145/3449198

[J9]. **Tsai, C. H.**, Rodriguez G. R., Li, N., Robert, J., Serpi, A. & Carroll, J.M. (2020). Experiencing the Transition to Remote Teaching and Learning during the COVID-19 Pandemic. Interaction Design and Architecture(s) Journal – IxD&A. https://doi.org/10.55612/s-5002-046-004

[J10]. Sandbulte, J., **Tsai, C. H.** & Carroll, J. M. (2020). Family's Health: Opportunities for Non-Collocated Intergenerational Families Collaboration on Healthy Living. *International Journal of Human-Computer Studies*. https://doi.org/10.1016/j.ijhcs.2020.102559

[J11]. **Tsai, C. H.**, & Brusilovsky, P. (2020). The Effects of Controllability and Explainability in a Social Recommender System. User Modeling and User-Adapted Interaction. https://doi.org/10.1007/s11257-020-09281-5

[J12]. **Tsai, C. H.**, & Brusilovsky, P. (2019). Exploring Social Recommendations with Visual Diversity-Promoting Interfaces. *ACM Transactions on Interactive Intelligent Systems (TiiS)*, 10(1), 1-34. https://doi.org/10.1145/3231465

[J13]. Amal, S., **Tsai, C. H.**, Brusilovsky, P., Kuflik, T., & Minkov, E. (2019). Relational social recommendation: Application to the academic domain. *Expert Systems with Applications*, 124, 182-195. https://doi.org/10.1016/j.eswa.2019.01.061

Peer-reviewed Conference Papers

[C1]. Mirzac, V., Thallemer, A., Yu, M. C., Chen, Y. C. & **Tsai, C. H.** (2025). Evaluating the Use of AI Agents in Teaching Doctor-Patient Communication Skills. In International Conference on Disruptive Technologies, Tech Ethics and Artificial Intelligence (DiTTEt'25). Cham: Springer Nature Switzerland.

[C2]. Osamau, E. & **Tsai, C. H.** (2025). Generative AI in Policy-Informed Insurance Adjudication: A Comparative Analysis. In International Conference on Disruptive Technologies, Tech Ethics and Artificial Intelligence (DiTTEt'25). Cham: Springer Nature Switzerland.

[C3]. Chen, Y. C., **Tsai, C. H.** & Zendejas, E. (2025). Multi-level Collaborative Governance Framework for Designing Accountable AI Systems for Emergency Management. In Proceedings of the 26th Annual International Conference on Digital Government Research. (dg.o'25)

[C4]. Ashaduzzaman, M., Nguyen, T. & **Tsai, C. H.** (2024). Explaining Social Recommendations Using Large Language Models. In International Conference on Disruptive Technologies, Tech Ethics and Artificial Intelligence (DiTTEt'24). Cham: Springer Nature Switzerland.

[C5]. Bano, T., Vadapalli, J., Karki, B., Thoene, M., Vanormer, M., Anderson Berry, A. & **Tsai, C. H.** (2024). Utilizing Retrieval-Augmented Large Language Models for Pregnancy Nutrition Advice. In International Conference on Disruptive Technologies, Tech Ethics and Artificial Intelligence (DiTTEt'24). Cham: Springer Nature Switzerland.

[C6]. Karki, B., Zhong, X., **Tsai, C. H.**, & Huang, P. C. (2024). Deep Learning-based Text-in-Image Watermarking. In 2024 IEEE conference on multimedia information processing and retrieval (MIPR'24).

[C7]. **Tsai, C. H.**, Nandy, G., House, D., & Carroll, J.M. (2024). Ensuring Transparency in Using ChatGPT for Public Sentiment Analysis. In Proceedings of the 25th Annual International Conference on Digital Government Research. (dg.o'24)

[C8]. Wang, Y. F., Chen, Y. C., Huang, Y. C., Redwing, C., & **Tsai, C. H.** (2024). Tribal Knowledge Cocreation in Generative Artificial Intelligence Systems. In Proceedings of the 25th Annual International Conference on Digital Government Research. (dg.o'24)

[C9]. Karki, B., Zhong, X., Chen, Y. T. and **Tsai, C. H.** (2024). Interpretable Classification of Myositis from Muscle Ultrasound Images. Proceedings of the 2024 8th International Conference on Medical and Health Informatics (ICMHI 2024).

[C10]. **Tsai, C. H.**, Kadire, S., Sreeramdas, T., Vanormer, M., Thoene, M., Hanson, C., Anderson Berry, A. & Khazanchi, D. (2023). Generating Personalized Pregnancy Nutrition Recommendations with GPT-Powered AI Chatbot. In: 20th International Conference on Information Systems for Crisis Response and Management (ISCRAM 2023).

[C11]. **Tsai, C. H.**, Rayi, P., Kadire, S., Wang, Y. F., Krafka, S., Zendejas, E., and Chen, Y. C. (2023). Co-Design Disaster Management Chatbot with Indigenous Communities. In: 20th International Conference on Information Systems for Crisis Response and Management (ISCRAM 2023).

[C12]. Li, N., Rodriguez G. R., Xu Y., Bhatt P., Nguyen H., Serpi, A., **Tsai, C. H.** & Carroll, J.M. (2022). Picturing One's Self: Camera Use in Zoom Classes during the COVID-19 Pandemic. In Proceedings of the Seventh ACM Conference on Learning@Scale. (**Best Paper Award**)

[C13]. Jo, J., Knearem, T., **Tsai, C. H.**, & Carroll, J. M. (2022). Using Local Community to Ease Long Haul Uncertainty during the COVID-19 Pandemic. Proceedings of International Conference on Information (iConference).

[C14]. Gui, F., **Tsai, C. H.**, & Carroll, J. M. (2022) Community acknowledgment: engaging community members in volunteer acknowledgment. Proceedings of the ACM on Human-Computer Interaction (GROUP).

[C15]. Knearem, T., Jo, J., **Tsai, C. H.**, & Carroll, J. M. (2022) Making Community Beliefs and Capacities Visible Through Care-mongering During COVID-19. Proceedings of the ACM on Human-Computer Interaction (GROUP).

[C16]. Knearem, T., Jo, J., **Tsai, C. H.**, & Carroll, J. M. (2021). Making Space for Support: An Exploratory Analysis of Pandemic-Response Mutual Aid Platforms. Proceedings of the 10th international conference on communities & technologies-transforming communities.

[C17]. Jo, J., Knearem, T., **Tsai, C. H.**, & Carroll, J. M. (2021). COVID Kindness: Patterns of Neighborly Cooperation during a Global Pandemic. Proceedings of the 10th international conference on communities & technologies-transforming communities. (**Best Paper Award**)

[C18]. **Tsai, C. H.**, You, Y., Gui, X., Kou, Y. & Carroll, J. M. (2021). Exploring and Promoting Diagnostic Transparency and Explainability in Online Symptom Checkers. *2021 CHI Conference on Human Factors in Computing Systems*.

[C19]. Lee, S., Reddie, M., **Tsai, C. H.**, Beck, J., Rosson M. B., & Carroll, J. M. (2020). The Emerging Professional Practice of Remote Sighted Assistance for People with Visual Impairments. *In Proceedings of the 2020 CHI Conference on Human Factors in Computing Systems*.

[C20]. **Tsai, C. H.**, & Brusilovsky, P. (2019). Evaluating visual explanations for similarity-based recommendations: User perception and performance. *In Proceedings of the 27th ACM Conference on User Modeling, Adaptation and Personalization* (pp. 22-30).

[C21]. **Tsai, C. H.**, & Brusilovsky, P. (2019). Explaining recommendations in an interactive hybrid social recommender. *In Proceedings of the 24th International Conference on Intelligent User Interfaces* (pp. 391-396).

[C22]. **Tsai, C. H.**, & Brusilovsky, P. (2018). Beyond the ranked list: User-driven exploration and diversification of social recommendation. *In 23rd International Conference on Intelligent User Interfaces* (pp. 239-250). (**Best Student Paper Honorable Mention Award**)

[C23]. **Tsai, C. H.**, & Brusilovsky, P. (2017). Leveraging interfaces to improve recommendation diversity. *In Adjunct Publication of the 25th Conference on User Modeling, Adaptation and Personalization* (pp. 65-70).

[C24]. **Tsai, C. H.**, & Brusilovsky, P. (2017). Providing control and transparency in a social recommender system for academic conferences. *In Proceedings of the 25th Conference on User Modeling, Adaptation and Personalization* (pp. 313-317).

[C25]. **Tsai, C. H.** (2016). A Fuzzy-based Personalized Recommender System for Local Businesses. *In Proceedings of the 27th ACM Conference on Hypertext and Social Media* (pp. 297-302).

Peer-reviewed Posters & Extended Abstracts & Doctoral Colloquium Papers

[P1]. Chen, W., Le, A., Athota, P., Ashaduzzaman, M. & **Tsai, C. H.** (2025). Leveraging Conversational AI for Adolescent Medical Financial Education. In Proceedings of the 26th Annual International Conference on Digital Government Research. (dg.o'25)

[P2]. Vadapalli, J., Gupta, S., Karki, B., & **Tsai, C. H.** (2024). Incorporating Citizen-Generated Data into Large Language Models. In Proceedings of the 25th Annual International Conference on Digital Government Research. (dg.o'24)

[P3]. Chandrasekar, H., Gupta, S., Liu, C. T., & **Tsai, C. H.** (2024). Leveraging Large Language Models for Effective Organizational Navigation. In Proceedings of the 25th Annual International Conference on Digital Government Research. (dg.o'24)

[P4]. Gupta, S., Chen, Y. C., & **Tsai, C. H.** (2024). Utilizing Large Language Models in Tribal Emergency Management. *In Proceedings of the 29th International Conference on Intelligent User Interfaces Companion (IUI'24)*.

[P5]. Rahdari, B., **Tsai, C. H.**, & Brusilovsky, P. (2019). Expanding Controllability of Hybrid Recommender Systems: From Positive to Negative Relevance. *In The Thirty-Second International Flairs Conference*. (**Runner-up for Best Poster Award**)

[P6]. **Tsai, C. H.** (2018). Diversity-Enhanced Recommendation Interface and Evaluation. Diversity-Enhanced Recommendation Interface and Evaluation. *In Proceedings of the 2018 Conference on Human Information Interaction & Retrieval (CHIIR'18)*.

[P7]. **Tsai, C. H.** & Brusilovsky, P. (2018). Explaining social recommendations to casual users: Design principles and opportunities. *In Proceedings of the 23rd International Conference on Intelligent User Interfaces Companion (IUI'18)* (pp. 1-2).

[P8]. **Tsai, C. H.** (2017). An interactive and interpretable interface for diversity in recommender systems. *In Proceedings of the 22nd International Conference on Intelligent User Interfaces Companion (IUI'17)* (pp. 225-228).

[P9]. **Tsai, C. H.** & Brusilovsky, P. (2016). A personalized people recommender system using global search approach. *IConference 2016 Proceedings*.

[P10]. Champagne, R., Guerra, J., **Tsai, C. H.**, Monahan, J., & Farzan, R. (2015). Fuzziness in LGBT non-profit ICT use. *In Proceedings of the Seventh International Conference on Information and Communication Technologies and Development* (pp. 1-4).

[P11]. **Tsai, C. H.** & Lin, Y. R. (2015). The evolution of scientific productivity of junior scholars. *IConference 2015 Proceedings*.

[P12]. **Tsai, C. H.** & Lin, Y. R. (2014). From media reporting to international relations: a case study of asia-pacific economic cooperation (APEC). *In Proceedings of the 2014 ACM conference on Web science* (pp. 279-280).

Peer-reviewed Workshop Papers

[W1]. Bano, T., Gupta, S., Chen, Y. C., Zendejas, E., Krafka, S. & **Tsai, C. H.**. (2025). Exploring Responsible Use of Generative AI in Disaster Resilience for Indigenous Communities. *In Adjunct Publication of the 33rd Conference on User Modeling, Adaptation and Personalization (LLM4GOOD)*.

[W2]. Ashaduzzaman, M. & **Tsai, C. H.**. (2025). Leveraging Generative AI to Improve Comprehensibility in Social Recommender Systems. *In Adjunct Publication of the 33rd Conference on User Modeling, Adaptation and Personalization (ExUM)*.

[W3]. Nandy, G., Gupta, S., Afzali, F. M., Peeples, E., Pilon, B. & **Tsai, C. H.**. (2025). Balancing Health Information-Seeking Through Retrieval-Augmented Generation-Based LLM Chatbot. *In Adjunct Publication of the 33rd Conference on User Modeling, Adaptation and Personalization (FairUMAP)*.

[W4]. Sreeramdas, T., Chandrasekar, H., Gudibandi, L. & **Tsai, C. H.**. (2023). Co-Design for Indigeneity and Sovereignty. *Indigeneity @ CSCW 2023*

[W5]. **Tsai, C. H.** & Brusilovsky, P. (2020). User Feedback in Controllable and Explainable Social Recommender Systems: a Linguistic Analysis. *In CEUR Workshop Proceedings (IntRS '20)*

[W6]. **Tsai, C. H.**, Huhtamäki, J., Olsson, T., & Brusilovsky, P. (2020). Diversity Exposure in Social Recommender Systems: A Social Capital Theory Perspective. *In CEUR Workshop Proceedings (IntRS '20)*

[W7]. Carroll, J. M., **Tsai, C. H.**, & Lee S. (2020). Talk the Talk: How Human Conversational Agents Build Trust, In The CHI 2020 Workshop of Conversational Agents for Health and Wellbeing.

[W8]. **Tsai, C. H.**, Brusilovsky, P., & Rahdari, B. (2019). Exploring User-Controlled Hybrid Recommendation in a Conference Context. *IUI Workshops (ESIDA '19)*.

[W9]. **Tsai, C. H.**, & Brusilovsky, P. (2019). Designing Explanation Interfaces for Transparency and Beyond. *IUI Workshops (ATEC '19)*.

[W10]. **Tsai, C. H.** & Brusilovsky, P. (2017). Enhancing recommendation diversity through a dual recommendation interface. *In CEUR Workshop Proceedings (IntRS "17)* (Vol. 1884, pp. 10-15).

[W11]. **Tsai, C. H.** & Lin, Y. R. (2016). Tracing and predicting collaboration for junior scholars. *In Proceedings of the 25th International Conference Companion on World Wide Web* (pp. 375-380).

Book Chapter

[B1]. **Tsai, C. H.** & Carroll, J. M. (2022). Logic and Pragmatics in AI Explanation. xxAI - Beyond explainable Artificial Intelligence. Springer, Cham.

[B2]. Carroll, J. M., Romera Rodriguez, G., Li, N., & **Tsai, C. H.** (2021). Cultivating and Leveraging Continuous Accountability Through Mundane Infrastructures for Critical Thinking. In Perspectives on Digitally-Mediated Team Learning (pp. 19-32). Springer, Cham.

Dissertation/Thesis

[T1]. **Tsai, C. H.** (2020). Controllability and explainability in a hybrid social recommender system (PhD dissertation, University of Pittsburgh).

[T2]. **Tsai, C. H.** (2008). A study of cloud computing SOA service platform architecture and operation mechanism (Master's thesis, National Chengchi University).

Grants

Federal/External Funding

- PI. **Chun-Hua Tsai**. Image-Based Nutrition Guidance: Developing a Multimodal AI System for Personalized Dietary Recommendations. USD $5,000. AIM-AHEAD – North/Midwest Hub, Duke Institute for Health Innovations Development Award. (Grant Period: 2025).
- PI. **Chun-Hua Tsai**. Using Multimodal AI to Enhance Prenatal Nutrition: A Machine Learning Approach to Analyze Food Images and Provide Personalized Dietary Advice. USD $30,000. The Great Plains IDeA-CTR Network: Biostatistics, Epidemiology, and Data Science Methodology Pilot Grant. (Grant Period: 2025-2026).

- PI. Yu-Che Chen. Co-PI. **Chun-Hua Tsai** (40% Contribution), Edouardo Zendejas. DASS: A Multi-level Collaborative Design Framework for Cross-sovereignty Software Accountability. USD $749,232. National Science Foundation: CISE. (Grant Period: 2023-2026).
- PI. **Chun-Hua Tsai**. CRII: HCC: REU: Empowering Human-AI Collaboration through Conversational Explanations. USD $182,293. National Science Foundation: CISE. (Grant Period: 2022-2025).

State/NU Funding

- PI. **Chun-Hua Tsai**. Co-PI. Ann Anderson Berry. Using AI-powered Chatbots to Enhance Health Equity and Patient-Centered Care for Low Socioeconomic Status Communities. USD $149,783. Nebraska University (NU) Collaboration Initiative. (Grant Period: 2023-2025).
- PI. **Chun-Hua Tsai**. Co-PI. Deepak Khazanchi, Ann Anderson Berry, Corrine K Hanson and Melissa Thoene. Promoting Optimal Nutrition in Pregnant Women through AI-Based Interactive Smart Chatbots. USD $7,494. Nebraska University (NU) Collaboration Initiative. (Grant Period: 2022-2023).

Internal Funding

- PI. **Chun-Hua Tsai**. AI Course Development Grant. USD $2,000. Division of Innovative and Learning-Centric Initiatives, , University of Nebraska Omaha. (Grant Period: 2024-2025).
- PI. Xin Zhong. CO-PI. Harvey Siy, **Chun-Hua Tsai**, Zhihao Yu, Xiaoyue Zoe Cheng and Victor Huang. AI CodeLab System: Prototype Development and Pilot Study for Cross-College AI Programming Education at UNO. USD $164,268. Weitz Innovation and Excellence Fund. University of Nebraska Omaha. (Grant Period: 2025-2027).
- PI. Mahadevan Subramaniam. CO-PI. Victor Winter, Prashanti Manda, **Chun-Hua Tsai**, Joel Elson and Derrick Nero. AI-CCORE: Artificial Intelligence Center for Collaborative Outreach Research and Education. USD $998,238.46. Weitz Innovation and Excellence Fund. University of Nebraska Omaha. (Grant Period: 2025-2030).
- PI. **Chun-Hua Tsai**. Leveraging Large Language Models for Effective Organizational Navigation. USD $10,000. College of Information Science & Technology. University of Nebraska Omaha. (Grant Period: 2024-2025).
- PI. **Chun-Hua Tsai**. Classification of Myositisfrom Muscle Ultrasound Images using Machine Learning. USD $10,000. The University Committee on Research and Creative Activity (UCRCA), University of Nebraska Omaha. (Grant Period: 2023-2024).
- PI. **Chun-Hua Tsai**. Promoting Community Health Access through Explainable AI-Mediated Communication. USD $75,000. Office of Research and Creative Activity (ORCA), University of Nebraska Omaha. (Grant Period: 2021-2023).
- PI. **Chun-Hua Tsai**. Segmentation of Lecture Videos Towards Customized Learning. USD $750. The University Committee for the Advancement of Teaching (UCAT), University of Nebraska Omaha. (Grant Period: 2021-2022).
- PI. **Chun-Hua Tsai**. Data Quality Analysts for Misinformation Projects. USD $10,000. College of IST, Penn State University. (supported two undergraduate students for summer internship) (Grant Period: 2020).
- Senior Personnel. **Chun-Hua Tsai**. Collaborators. Peter Brusilovsky, Eliza Beth Littleton, Dmitriy Babichenko, Lorin Grieve, Hung Chau, Jordan Ariel Barria Pineda, Shreya Shenoy, Kar-Hai Chu, Zara Blum, Ravi Patel. (2018). "It's Who You Know": Feasibility of a Hybrid Recommender System to Connect Students with Informal Social Networks of Pitt Researchers. USD $10,000. Provost's Personalized Education Grant, University of Pittsburgh. (Grant Period: 2018-2019).

Industry Funding and Gifts

- CloudBank: Managed Services to Simplify Cloud Access for Computer Science Research and Education, $10,000 Cloud Credits AWS Cloud Credits, (Grant Period: 2023-2024).
- Google Cloud Research Credits, $5,000 Cloud Credits. (Grant Period: 2022-2023).

Teaching Experience

Courses As Instructor:

- CSCI 4250: Human Computer Interaction, Undergraduate Course, Department of Computer Science, University of Nebraska at Omaha (Fall 2025)
- ISQA 8310: Information Technology Infrastructure & Cloud, Graduate Course, Department of Information Systems & Quantitative Analysis (ISQA), University of Nebraska at Omaha (Spring 2024; Spring 2025)
- ISQA 8016/4010: Business Intelligence, Graduate Course, Department of Information Systems & Quantitative Analysis (ISQA), University of Nebraska at Omaha (Fall 2023; Fall 2024; Fall 2025)
- ISQA 3310: Managing Database Environment, Undergraduate Course, Department of Information Systems & Quantitative Analysis (ISQA), University of Nebraska at Omaha (Fall 2022; Fall 2024)
- ISQA 8380: Enterprise Architecture and Systems Integration, Graduate Course, Department of Information Systems & Quantitative Analysis (ISQA), University of Nebraska at Omaha (Spring 2022; Summer 2022; Spring 2023; Spring 2024; Spring 2025)
- EMIT 8700: Emerging Challenges, Graduate Course, Executive Master of Science in Information Technology (EMIT), University of Nebraska at Omaha (Summer 2022)
- ISQA 3400: Information Technology Infrastructure, Undergraduate Course, Department of Information Systems & Quantitative Analysis (ISQA), University of Nebraska at Omaha (Fall 2021; Spring 2022; Spring 2023; Fall 2023)
- IST 311: Object-Oriented Design and Software Applications, Undergraduate Course, College of Information Sciences and Technology, The Pennsylvania State University (Spring 2020; Spring 2021) (hybrid teaching due to COVID-19 pandemic)
- INFSCI 2710: Database Management, Graduate Course, School of Computing and Information, University of Pittsburgh (Spring 2018; Summer 2018; Fall 2018; Spring 2019)
- INFSCI 2300: Human Information Processing, Graduate Course, School of Computing and Information, University of Pittsburgh (Summer 2017; Fall 2017)

Courses As Teaching Assistant:

- INFSCI 2730: E-Business, Graduate Course, School of Information Sciences, University of Pittsburgh (Spring 2017)
- INFSCI 2560: Web Technologies & Standards, Graduate Course, School of Information Sciences, University of Pittsburgh (Fall 2016; Spring 2017)

Professional Services

- Grant Reviewer
  - NSF Panel Reviewer (2025-1, 2025-2)
  - UNMC Panel Reviewer (2025)
- Program Committee
  - ACM RecSys (2024); CSCW (2022-2025); Hypertext (2020-2021, 2025); UMAP (2019-2025); DITTET (2024-2025); ECIR (2024-2025)
  - ACM CHI-LBR (2021-2023); CHI-EmpathiCH (2022); UMAP-LBR (2019-2023); UMAP-Doctoral Consortium (2022); UMAP-ExUM (2022); RecSys-IntRS (2020-2024); IUI-ESIDA (2019; 2021); IUI-ExUM (2022); IUI-Poster&Demo (2022); WSDM-IDM (2019)
- Journal Reviewer
  - Social Sciences & Humanities Open (2025)

- Computers in Human Behavior: Artificial Humans (2025)
- Information Processing and Management (2025)
- JMIR Formative Research (2025)
- Discover Education (2024)
- Discover Public Health (2024)
- Discover Global Societ (2024)
- International Journal of Human-Computer Studies (2024,2025-1,2025-2)
- PeerJ Computer Science (2024)
- Discover Education (2024)
- The Collegiate Aviation Review-International (CARI) (2024)
- Scientific Reports - Nature (2024)
- The Journal of Information Technology (JIT) (2023)
- International Journal of Human-Computer Interaction (IJHCI) (2022; 2023)
- The IEEE Transactions on Neural Networks and Learning Systems (2022)
- Information Sciences (2022)
- Information Processing & Management (2022)
- *ACM Transactions on Information Systems* (2022)
- *ACM Transactions on Interactive Intelligent Systems* (2020; 2022; 2023; 2024)
- *International Journal on Digital Libraries* (2020)
- *The Journal of Management Analytics (JMA)* (2020)
- *User Modeling and User-Adapted Interaction (UMUAI)* (2019; 2021)
- *IEEE Transactions on Knowledge and Data Engineering (TKDE)* (2019)
- *World Wide Web: Internet and Web Information Systems (WWW)* (2019)
- *Behaviour & Information Technology (BIT)* (2019, 2024)
- *Paladyn, Journal of Behavioral Robotics* (2019)
- *Big Data* (2018)
- KSII *Transactions on Internet and Information Systems* (2017)

- Conference Reviewer
  - ISCRAM (2023-2025); HICSS (2022); ACM IMWUT (2022); AMCIS (2022); AIS ECIS (2022); ACM IUI (2022-2025); ACM CHI Play (2021); ACM CSCW (2021-2022); ACM CHI (2021-2025); iConference (2019-2025); SBP-BRiMS (2019); UMAP (2016); AAAI ICWSM (2015); ACM Web Sci (2014-2015); ACM WWW (2014-2016)

Honors and Awards

- Best Paper Award, ACM Conference on Learning @ Scale (L@S), 2022
- Best Paper Award, International Conference on Communities and Technologies (C&T), 2021
- Outstanding PC Members, ACM Conference on User Modeling, Adaptation and Personalization (UMAP), 2019
- Runner-up for Best Poster Award, International Flairs Conference, 2019

- Best Student Paper Honorable Mention Award, ACM International Conference on Intelligent User Interfaces (IUI), 2018
- GPSG Leadership & Service Award, University of Pittsburgh, 2017
- People's Choice Award, Three Minute Thesis, University of Pittsburgh, 2016
- 2nd Place, iFest 2015 Digital Poster Competition, University of Pittsburgh, 2015
- 2nd Place, International ICT Innovative Services Contest, Taiwan, 2007

Fellowships

- UNO '24 TANDEM Community Mentorship Training Stipend, USD $100
- UNO '23 TANDEM Community Mentorship Training Stipend, USD $250
- CSST '22 / CSST Summer Institute Travel Grant, USD $2,000
- CHI '22 / SIGCHI Childcare Travel Grants, 2022, USD $1,000
- CHI '21 / SIGCHI Gary Marsden Travel Award, 2021, USD $470
- IUI '19 / SIGCHI Student Travel Grant, 2019, USD $1,800
- CHIIR '18 / ACM Student Travel Grants, 2018, USD $2,000
- IUI '18 / NSF Student Travel Grants, 2018, USD $1,400
- RecSys '17 Student Travel Grants, 2017, USD $1,300
- UMAP '17 / NSF Student Travel Grants, 2017, USD $1,840
- IUI '17 / NSF Student Travel Grants, 2017, USD $1,550
- HT '16 / SIGWEB Student Travel Grants, 2016, USD $1,500
- WWW '16 Student Travel Grants, 2016, USD $600
- iConf '16 Student Travel Grants, 2016, USD $900
- iConf '15 Student Travel Grants, 2015, USD $1,000
- ICTD '15 Student Travel Grants, 2015, USD $850
- INSNA '15 Student Travel Grants, 2015, USD $350
- WebSci '14 / SIGWEB Student Travel Grants, 2014, USD $600
- Taiwan Government Fellowship for Studying Abroad, 2013-2016, USD $150,000

Academic Advising

Postdoc Mentor

- Srishti Gupta (2023-2024)

PhD Advisor

- Taranum Bano, College of IS&T, University of Nebraska at Omaha (2024-)
- Md Ashaduzzaman, College of IS&T, University of Nebraska at Omaha (2024-)
- Gargi Nandy, College of IS&T, University of Nebraska at Omaha (2023-)

Doctoral Dissertation Committee

- Andrew Burk, School of Public Administration, University of Nebraska at Omaha. (2024-)
- Anoop Mishra, College of IS&T, University of Nebraska at Omaha. Dissertation: *Assessing Perceived Fairness in Machine Learning (ML) Process.* (2023-)

- Yi-Fan Wang, School of Public Administration, University of Nebraska at Omaha. Dissertation: *Artificial Intelligence System Codesign with Native American Communities for Tribal Emergency Management*. (2022-2024)
- Yue You, College of IST, Penn State University. Dissertation: *Evaluating and Designing Symptom Checkers: A User-Centered Approach*. (2021-2024)

Master's Thesis Committee

- Penjani Banda, College of Communication, University of Nebraska at Omaha, *Thesis: Integrating Generative Ai In Higher Education: A Case Study On Communication Education At The University Of Nebraska* (2024-2025)
- Bhavani Malisetty, Computer Science, College of IS&T, University of Nebraska at Omaha, *Thesis: Improving The Understanding Of Privacy Policies Using Large Language Models (LLMs)* (2024-2025)
- Bishwa Karki, Computer Science, College of IS&T, University of Nebraska at Omaha, *Thesis: Watermarking outputs of large language model*. (2023-2024)
- Alex Borchers, Computer Science, College of IS&T, University of Nebraska at Omaha, *Thesis: Metamorphic Testing and Mutation Testing for SVM Image Classification*. (2022-2023)
- Shane Schutte, IT Innovation, College of IS&T, University of Nebraska at Omaha, *Thesis: Tenant Sentiment Effects of Smart Home Technology in Short-Term Rentals*. (2021-2023)
- Yanting Wu, College of IST, Penn State University, *Thesis: Role of Videoconferencing Affordances in Remote Education during COVID-19*. (2021)

Graduate Research

- Himajasubhamrutha Sammeta, College of IS&T, University of Nebraska at Omaha (2025-)
- Emmanuel Osamau, Department of Mathematics, University of Nebraska at Omaha (2024-)
- Val Mirzac, College of IS&T, University of Nebraska at Omaha (2024-)
- Jagadeesh Vadapalli, College of IS&T, University of Nebraska at Omaha (2023-2024)
- Vasanth Patel Andela, College of IS&T, University of Nebraska at Omaha (2023-2024)
- Akhila Chevula, College of IS&T, University of Nebraska at Omaha (2023-2024)
- Lavanya Gudibandi, College of IS&T, University of Nebraska at Omaha (2023-2024)
- Sri Sai Tejesvi Sreeramdas, College of IS&T, University of Nebraska at Omaha (2022-2024)
- Sathvik Kadire, College of IS&T, University of Nebraska at Omaha (2022-2024)
- Koushik Reddy Pasula, College of IS&T, University of Nebraska at Omaha (2022-2023)
- Paul Rayi, College of IS&T, University of Nebraska at Omaha (2022-2023)
- Mohammad Bakhtiari, College of IS&T, University of Nebraska at Omaha (2022-2023)
- Raghava Sai Kowlasker, College of IS&T, University of Nebraska at Omaha (2021-2022)
- Jingyi Xie, College of IST, Penn State University (2020-2022)
- Na Li, College of IST, Penn State University (2019-2022)
- Tiffany Knearem, College of IST, Penn State University (2019-2022)
- Fanlu Gui, College of IST, Penn State University (2019-2022)
- Srishti Gupta, College of IST, Penn State University (2019-2022)
- Jeongwon Jo, College of IST, Penn State University (2019-2022)
- Romera Rodriguez Guillermo, College of IST, Penn State University (2019-2022)
- Sameer Sapre, Eberly College of Science, Penn State University (2020-2021)

- Jomara Sandbulte (Binda), College of IST, Penn State University (2019-2021)
- Sooyeon Lee, College of IST, Penn State University (2019-2021)

Undergraduate Research

- Thi Nguyen, Computer Science, University of Nebraska at Omaha (NSF REU) (2024-)
- Corice Lieb, Tribal Emergency Management, University of Nebraska at Omaha (NSF REU) (2024-)
- Haresh Chandrasekar, Computer Science, University of Nebraska at Omaha (2023-2024)
- Nicholas Alico, College of IST, Penn State University (2020-2021)
- Debanshu Das, College of IST, Penn State University (2020)
- Kairui Wang, College of IST, Penn State University (2020)

University Services

- Committee Member, The UNO Asian American Pacific Islander (AAPI) Faculty-Staff Affinity Group, the University of Nebraska at Omaha (2023-).
- Committee Member, Doctoral Program Committee (DPC), College of Information Science & Technology, the University of Nebraska at Omaha (2022-).
- Committee Member, Graduate Program Committee (GPC), Department of Information Systems and Quantitative Analysis (ISQA), the University of Nebraska at Omaha. (2022-).
- Hiring Committee, Committee Member, Department of Information Systems and Quantitative Analysis (ISQA), the University of Nebraska at Omaha. (2022-2023).

Invited Talks / Panelist / Presentations

- Gupta, S., Nandy, G., Afzali, F. M., Peeples, E., Pilon, E. & **Tsai, C. H.** (2024). An Exploratory Study of HopeCare: A Generative AI-Powered Chatbot Supporting Caregivers of Children with Hypoxic-Ischemic Encephalopathy. The Child Health Research Institute Annual Scientific Conference.
- Nandy, G., Gupta, S., Afzali, F. M., Peeples, E., Pilon, E. & **Tsai, C. H.** (2024). Enhancing Caregiver Information Seeking with Retrieval-Augmented Generation: A Case Study of Neonatal Hypoxic-Ischemic Encephalopathy. The Child Health Research Institute Annual Scientific Conference.
- Panelist, AI and Machine Learning @ Indigenous Computational Futures (2025)
- Panelist, UR2PhD: Panel During Undergraduate Research Methods Course (2024)
- Oral Presenter (with Dr. Yu-Che Chen), Achieving AI System Accountability for Collaborative Emergency Management between Tribal Nations and U.S. Governments, Citizen Efficacy through New Technologies, American Society for Public Administration (ASPA) Annual Conference (2024)
- Panelist, "Evaluating explanations: How to evaluate the quality of explanations through both subjective and objective measures?", 2nd Workshop on Measuring the Quality of Explanations in Recommender Systems (2023)
- Oral Presenter (with Dr. Yu-Che Chen), "AI Chatbots and their Government Applications", Nebraska City/County Managers Association Summer Conference, Omaha, NE (2023).
- Oral Presenter, "AI Chatbots and their Government Applications", UNO CIS&T Engagement with MITRE, Omaha, NE (2023).
- Discussant & Presenter, Session of Technology & Well-being, The 18th Annual Big XII+ MIS Research Symposium, University of Houston (2022).
- Oral Presenter, "Facilitating and Empowering Human-AI Collaboration through Control and Explanations", Bite Sized Research @ College of Information Science and Technology (IS&T) at University of Nebraska at Omaha (2022).

- Moderator, Student Research and Creative Activity Fair, Office of Research and Creative Activity at University of Nebraska at Omaha (2022).
- Oral Presenter, "Facilitating and Empowering Human-AI Collaboration through Control and Explanations", Department of Information Sciences and Technology at George Mason University (2021).
- Oral Presenter, "Facilitating and Empowering Human-AI Collaboration through Control and Explanations", Department of Data Science and Knowledge Engineering (DKE) at University of Maastricht (2021).
- Oral Presenter, "Facilitating and Empowering Human-AI Collaboration through Control and Explanations", College of Engineering and Information Technology at University of Maryland Baltimore County (2021).
- Oral Presenter, "Facilitating and Empowering Human-AI Collaboration through Control and Explanations", Department of Computer and Information Technology (CIT) at Purdue University (2021).
- Oral Presenter, "It's Who You Know": A Hybrid Recommender System to Connect Students with Informal Social Networks of Pitt Researchers, ACCelerate: ACC Smithsonian Creativity and Innovation Festival at National Museum of American History (2019)
- Poster Presenter, Diversity and Excellence in Mentoring and Advising, 2019 Mentoring and Advising Summit, University of Pittsburgh (2019)
- Panelist, Parenting Panel for Graduate/Professional Students, The Office of the Provost, University of Pittsburgh (2019)
- Oral Presenter, "Teaching Databases and SQL using Intelligent Tutoring Systems", School of Computing and Information at University of Pittsburgh (2019)
- Poster Presenter, A Personalized People Recommender System Using Global Search Approach, School's Board of Visitors (BOV) , University of Pittsburgh (2016)

Computational Skills

- Programming Languages: JAVA, ASP.NET (C#)
- Scripting Languages: R, Python, PHP, SQL & JavaScript
- Systems: Linux (Ubuntu/Fedora/Redhat), Windows and System Virtualization